Morphophonology

Class 1: Why does phonology care about morphological boundaries?

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The object of study

- What is 'morphophonology'?
- Why should we be interested in it?

Breaking this down

- What is phonology?
- What would distinguish a pattern as 'morphophonological'?

What phonology computes

Perhaps:

- Set of principles governing the distribution of sounds, such as:
- What sounds may (or may not) occur in a given context
- How sounds change in different contexts
    - "Sounds": combinations of feature values

Contexts

Stop us and ask questions whenever we use a term or concept that you'd like us to define or clarify!

- Simple starting assumption: combinations of feature values
- Rules: A → B / C __ D
  - A,B,C,D are feature matrices
- Constraints: *CAD, Ident([±F])
  - CAD and F are feature matrices
- Restriction to feature matrices ⇒ "phonetic"

What makes a rule/constraint "non-phonetic"?

Phonetic conditioning

- Voicing assimilation in Catalan
  - [-son] → [αvoi] / __ [+cons,αvoi]
  - Obstruents assimilate in voicing to a following consonant
  - Applies within words and across word boundaries
  - Voicing also before sonorants

(examples on the next slide)

Voicing assimilation in Catalan

Within words	ˈozlu	‘Oslo’
	əzˈnɔp	‘snob’
	ˈbizβə	‘bishop’
	əzˈɣlaj	‘startle’
Across word boundaries	kaz ˈlɔʒik	‘logical case’
	ˌbuz nukˈturn	‘night bus’
	ˌnaz βuˈnik	‘pretty nose’
	pɛz ˈɣran	‘big weight’

Morphological conditioning

- Voicing assimilation of /s/ in English
  - s → z / ˈV__[+cons,+voi]

Within words	ˈazloʊ	'Oslo'
	ˈgazlɪŋ	'gosling'
	ˈæznɹ̩	'Asner'
	ˈazwald	'Oswald'
	ˈnæzdæk	'NASDAQ
Across word boundaries	ˈkɹɔs loʊ	'cross low'
	ˈpæs nu	'pass new'
	ˈkɹɔs wɔk	'cross walk'
	ˈbʌs deɪ	'bus day'

Examples abound

- Across the board
  - Cat./Span.: voicing assimilation, spirantization, r allophony ($ɾ \rightarrow r/\_\_]_{\sigma}$)
- Morphologically bounded
  - Obstruent devoicing in German, Catalan, etc.
  - Aspiration of voiceless stops in English
  - Flapping in English

Morphological contexts within the word

- Sensitivity to word boundaries
    - Warsaw Polish obstruent devoicing word-finally
    - But not at prefix+stem boundaries: *po*[d]-*adres* 'subaddress'
- Sensitivity to morpheme boundaries within the word
    - Catalan, German obstruent devoicing word-finally
    - Also at prefix+stem boundaries: Catalan: *sub*-V *su*[p]*adreça* ‘subaddress’

Parameterization

- Spanish s-aspiration in various dialects (Valiente 2025)

	$[_{St}$ __ + Suf.]	$[_{St}$ __]	$[_{Pref}$ __] # V	$[_{Wd}$ __] ##	$[_{Wd}$ __] ## V	$[_{Wd}$ __] ## C
Non-aspirating	arro[s]al	ca[s]pa	de[s]hecho	do[s]	ve[s] uno	ve[s] cuatro
Buenos Aires	arro[s]al	ca[h]pa	de[s]hecho	do[s]	ve[s] uno	ve[h] cuatro
Chilean	arro[s]al	ca[h]pa	de[s]hecho	do[∅]	ve[h] uno	ve[∅] cuatro
Granada	arro[s]al	ca[h]pa	de[h]hecho	do[h]	ve[h] uno	ve[h] cuatro
Río Negro	arro[s]al	ca[h]pa	de[s]hecho	do[h]	ve[h] uno	ve[h] cuatro

Table 1: /s/ weakening in Spanish

- Buenos Aires: aspiration before a consonant
- Granada: aspiration at prefix and word boundaries
- Río Negro: aspiration at word boundaries only

Even more fine-grained contexts

- Root vs. affix, prefix vs. suffix
- Classes of affixes
- Paradigmatic relations
- Phonologically exceptional morphemes
- Phonologically conditioned morphology
- …

Morphophonology

Why should we care?

- Of course, we have to account for the facts of phonological systems
- More generally: does this shed light on the computational nature of phonology?
- Marr's levels of analysis
  - Computational
  - Algorithmic
  - Implementation

What does phonology compute?

- A natural thought: "make sounds easy to produce and perceive"
- In order to do this, it regulates:
    - What sounds may (or may not) occur in a given context
    - How sounds change in different contexts

Marr's levels

Computational:

- What sounds may occur in a given context
- How sounds change in different contexts

Algorithmic:

- Markedness
- Faithfulness
- GEN, Ranking, etc.

Implementational: ???

Another possible hypothesis

Optimize accuracy of learning, producing, and recognizing morphemes by restricting sounds and sound combinations.

- Limited inventory of sounds, maximally distinct
- Minimize alternations
- Cues to segmentation

Our goals for this class

- Survey some ways in which phonology is sensitive to morphological structure
- Compare some approaches on the market
    - Hands-on analysis of case studies
- See what we can learn about phonology from the relative strengths and drawbacks of different approaches
- Learning from each other through discussion and debate!

Some mechanics

- Schedule of topics, slides, and links to readings will be posted on the following website:

  creteling2026.phonology.party

- Copyrighted PDF's will be posted on the CreteLing courses site
- Meet with us!

Our first "non-phonetic" context: boundaries

- Many phonological processes apply freely across word boundaries
  - Spirantization of voiced stops in Spanish
  - regressive voicing assimilation in Russian, Polish, Catalan, etc.
- But many do not
  - English /s/ voicing (*Oslo*), Catalan final devoicing (*sub-*), etc.

Aspiration in English

- Voiceless stops are aspirated in absolute syllable-initial position…
  - Word-initially: [pʰ]irate, [pʰ]ly, [pʰ]otato
  - Or before V́: su[pʰ]ose, com[pʰ]ile, su[pʰ]ly
- Unaspirated elsewhere
  - Medially before a stressless vowel: còm[p]ilátion
  - When not syllable-initial: s[p]írant, s[p]orádic, pas[t]él, as[t]óund
  - Assumption: VsTV syllabified V.sTV

Aspiration in English: OT

Some constraints (subset)

- $T^h/\_\,ˈV$
  - Voiceless stops must be aspirated before a stressed vowel
- $*sT^h$
  - No aspirated consonants after *s*
- $*C^h$
  - No aspirated consonants
- IO-Ident([±asp])
  - Corresponding underlying and surface should have same value of [±asp]

Aspiration in English: OT

/ˈhæpi/	*sTʰ	Tʰ/_ˈV	*Cʰ	IO-Ident([±asp])
a. ☞ hæpi				
b. hæpʰi			*!	*

/əˈpɛnd/	*sTʰ	Tʰ/_ˈV	*Cʰ	IO-Ident([±asp])
a. əpɛnd		*!		
b. ☞ əpʰɛnd			*	*

/əˈspaɪɹ/	*sTʰ	Tʰ/_ˈV	*Cʰ	IO-Ident([±asp])
a. ☞ əˈspaɪɹ		*		
b. əˈspʰaɪɹ	*!		*	*

Aspiration is predictable

…so the input value doesn't matter

/ˈhæp^{h}i/	*sT^h	T^h/_ˈV	*C^h	IO-Ident([±asp])
a. ☞ hæpi				*
b. hæp^{h}i			*!	

/əˈp^{h}ɛnd/	*sT^h	T^h/_ˈV	*C^h	IO-Ident([±asp])
a. əpɛnd		*!		*
b. ☞ əp^{h}ɛnd			*	

/əˈsp^{h}aɪɹ/	*sT^h	T^h/_ˈV	*C^h	IO-Ident([±asp])
a. ☞ əˈspaɪɹ		*		*
b. əˈsp^{h}aɪɹ	*!		*	

Word boundaries as a context

- English stop aspiration

sur[p^h]ríse	cu[p] ríse
no [k^h]úp	wo[k]e úp
at[t^h]íre	wha[t] íre
pas[t]el	pass [t^h]ell
dis[p]lay	press [p^h]lay

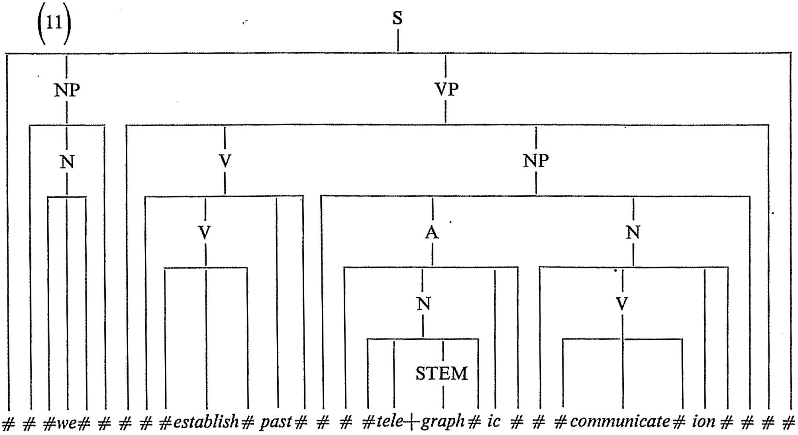
- Aspiration conditioned by context within the word
- Stress in next word, /s/ in previous word are irrelevant

Various approaches

- Allow rules/constraints to refer to boundaries, directly or indirectly (prosody)
- Allow phonology to evaluate pieces that are smaller than the entire phrase, and making the "word" one stage of evaluation (*cyclicity*)
- Allow phonology to enforce similarity between how strings are pronounced when part of a complex expression, and how they are pronounced in isolation (*correspondence*)

We'll explore each of these, in turn…

Boundaries as symbols in SPE



Boundaries as symbols

- Process applies only within words
  - $A \rightarrow B$ / C __ D (where C,D don't contain '#')
- Process applies only at or across boundaries
  - $A \rightarrow B$ / C __ #(D)
- Across the board (ATB): process applies **both** within and across boundaries
  - $A \rightarrow B$ / C(#) __ (#)D

A less stipulative approach?

- Hypothesis: prosody mediates between morphological structure and phonological processes
  - Phonology refers to prosodic structure
  - Morphological structure is tied to prosodic structure

Prosodic structure: prosodic hierarchy

(Selkirk 1984; Nespor and Vogel 1986)

Phonological utterance

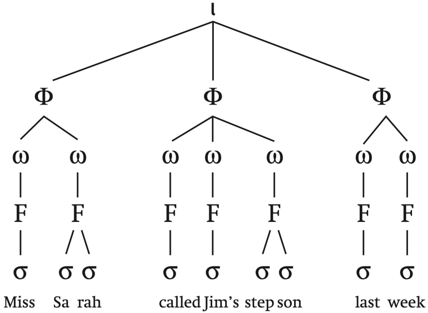
Intonational phrase (ι)

Phonological phrase (Φ)

Phonological word (ω)

Foot (F)

Syllable (σ)



Making prosody care about morphology

- Prosodic Morphology (McCarthy & Prince 1986)
  - Morphological units may be defined prosodically
- Alignment of morphological and prosodic domains

Applying this: English aspiration

sur[pʰ]ríse	cu[p] ríse
no [kʰ]úp	wo[k]e úp
at[tʰ]íre	wha[t] íre
pas[t]el	pass [tʰ]ell
dis[p]lay	press [pʰ]lay

- Prosodic restatement: aspiration within the ***prosodic*** *word*
- Morphological word boundaries mirrored by prosodic word boundaries

$[\text{surprise}]_{\text{PrWd}}$ vs. $[\text{cup}]_{\text{PrWd}}[\text{rise}]_{\text{PrWd}}$

Revised constraints to care about words

Some constraints (subset)

- T^h/_ˈV *within the word*
  - Voiceless stops must be aspirated before a stressed vowel
- $*sT^h$ *within the word*
  - No aspirated consonants after *s*
- $*C^h$
  - No aspirated consonants
- IO-Ident([±asp])
  - Corresponding underlying and surface should have same value of [±asp]

Now word boundaries matter

- *append* vs. *up-end*

/əˈpɛnd/	*sT^h	T^h/_ˈV	*C^h	IO-Ident([±asp])
a. əpɛnd		*!		
b. ☞ əp^{h}ɛnd			*	*

/ʌp ɛnd/	*sT^h	T^h/_ˈV	*C^h	IO-Ident([±asp])
a. ☞ [ʌp][ɛnd]				
b. [ʌp^h][ɛnd]			*!	*

Now word boundaries matter

- *aspire* vs. *less pie*

/əˈspaɪɹ/	*sTʰ	Tʰ/_ˈV	*Cʰ	IO-Ident([±asp])
a. ☞ əˈspaɪɹ		*		
b. əˈspʰaɪɹ	*!		*	*

/lɛs paɪ/	*sTʰ	Tʰ/_ˈV	*Cʰ	IO-Ident([±asp])
a. [lɛs][paɪ]		*!		
b. ☞ [lɛs][pʰaɪ]			*	*

Within the word

dis[p]lay	dis[p^h]lease
des[p]otism	dis[p^h]ossess
whis[p]ering	mis[p^h]erceive
pis[t]achio	mis[t^h]abulate
Pis[k]ataway	mis[k^h]ategorize

- Makes sense if stem initiates a word boundary
  - $[\text{di.splay}]_{\text{PrWd}}$
  - $\text{dis}[\text{please}]_{\text{PrWd}}$

Aligning prosodic and morphological boundaries

- McCarthy & Prince (1993) Generalized Alignment
  - Align(MCat, Edge, PCat, Edge)
- Example: Align(Stem, L, PrWd, L)
  - "Align-L" for short
  - Align(Stem, L, σ, L) also works here

Violated by $[\text{dɪs}(_{\text{stem}}\text{pliːz}]_{\text{PrWd}}$

Satisfied by $[\text{dɪs}]_{\text{PrWd}}[(_{\text{stem}}\text{pliːz}]_{\text{PrWd}}$

Alignment in English

/dɪs-pliːz/	Align-L	*sTʰ	Tʰ/_ˈV	*Cʰ	IO-Ident ([±asp])
a. [dɪsplˈiːz]	*!		*		
b. [dɪspʰlˈiːz]	*!	*		*	*
c. ☞ [dɪs][pʰlˈiːz]				*	*

- Recall: *sTʰ, Tʰ/_ˈV apply *within the word*

Thought question

- If we can circumvent violations of *sT^h by adding a word boundary, why don't we always add PrWd boundaries to license sT^h?
    - *aspire* [əs][pʰaɪɹ]

Another example: German

German (McCarthy and Prince 1993)

(a)	/juʁ-iʁ-ən/	[juʁiʁən]	'adjudicate'
	/pʁofɛsoːʁ/	[pʁofɛsoːɐ̯]	'professor'
	/holand/	[holant]	'Holland'
	/tsvaɪ-baɪnɪg/	[tsvaɪbaɪnɪç]	'two-legged'
	/egal/	[egal]	'indifferent'
(b)	/fɛʁ-iʁən/	[fɛʁʔiʁən]	'lose one's way' (verirren)
	/aʊf-ɛsən/	[aʊfʔɛsən]	'eat up' (aufessen)
	/tsol-amt/	[tsolʔamt]	'customs house' (Zollamt)
	/ʁaɪb-aɪzən/	[ʁaɪpʔaɪzən]	'grater' (Reibeisen)
	/bɛʁg-ab/	[bɛɐ̯kʔap]	'downhill' (Bergab)

Regular German syllabification

- Initial syllables require onsets

/amt/	ʔamt	'office'
/eɪzən/	ʔeɪzən	'iron'

- Medial /VCV/ syllabified V.CV

/pʁofɛsoːʁ/	pʁo.fɛ.soːɐ̯	'professor'
/tabuliːʁən/	ta.bu.liː.ʁən	'tabulate'

- Ranking: Onset >> Dep(ʔ)

Unexpected syllabification with prefixes

[ju.ʁiː.ʁən] *jurieren* vs. [fɛʁ.ʔiː.ʁən] *ver-irren*

[pʁo.ˈfɛ.soːɐ̯] *Professor* vs. [aʊfʔɛsən] *auf-essen*

McCarthy and Prince's insight for German (following others)

- As in English, prefix+stem looks like a word boundary
  - Glottal stop epenthesis for V-initial stems
  - Final devoicing of prefix
- Same as with compounds, which are more clearly word+word boundaries

Align-L?

- Recall the problem: desired winner [aʊf].[ʔ|ɛ.sən] is not actually aligned

/aʊf-ɛs-ən/	Onset	Align-L	Dep(ʔ)
a. [aʊf.ʔ|ɛ.sən]		*!***	*
b. [aʊ.f|ɛ.sən]		*!**	
c. [aʊf].[|ɛ.sən]	*!		
d. [aʊf].[ʔ|ɛ.sən]		*?	*
e. [aʊ].[f|ɛ.sən]		*	

Some options

- Fine-grained Align violations (ʔ < f)
- Align-R for the prefix?
  - Align(*auf*,R,PrWd,R)
  - How to generalize to Align(prefix,R,PrWd,R)?
- Give up on PrWd boundaries?
  - Or combine with one of the approaches to be discussed next time?

The prosodic status of the prefix

- Align-L demands PrWd boundary at beginning of stem, but what does that mean for the prefix?
- One possibility: separate PrWd
  - $[\text{aʊf}]_{\text{PrWd}}[\text{ʔɛsən}]_{\text{PrWd}}$
- Another possibility: adjunction
  - $[\text{aʊf}[\text{ʔɛsən}]_{\text{PrWd}}]_{\text{PrWd}}$
- Distinction may be relevant for other cases discussed later

Prefixes vs. suffixes

- Align-L analysis makes no commitment to…
  - Right edge of stem
  - Left edge of other categories (prefix, suffix)
- Stem+suffix boundaries do not show the same effects
  - pro[b]-ieren 'try', *pro[pʔ]ieren, Impe[d]-anz 'impedance', *Impe[tʔ]anz, legen[d]-är 'legendary', *legen[tʔ]är, Nu[d]-ismus 'nudism', *Nu[tʔ]ismus
  - Arguably compounds: -artig '-like', -äugig '-eyed'
- *Coda >> Align(Stem,R,PrWd,R)

Prefix+prefix combinations

- Prefix+prefix boundaries do seem to have ʔ epenthesis
  - ver[ʔ]ausgaben 'spend, exert'
- Requires a refinement of the Align-L constraint
  - More than just stems? or what is a stem?

And many more cases…

- Catalan prefixes
  - Like German, show "final" devoicing (*su*[p]*adreça* ‘subaddress’)
- Spanish s-aspiration
  - In prefixes, suffixes, etc.
- s-voicing in Catalan and Spanish
- We'll discuss some of these in greater detail in class 3 when we compare various approaches

Takeaway points: prosodic domains

- Effect of morphology mediated by prosody
  - Prosodic structure constrained to mirror morphological structure
  - But constraints may be outranked
- No particular prediction about which edges or types of constituents should show strongest effects, but there do seem to be recurring trends

The "goal" of phonology

- Prosodic boundaries aren't directly audible, so might seem pointless to require them
- Consequences of prosodic boundaries: stress, intonation, conditioning of segmental processes
- All of these are helpful in parsing/word recognition (Christophe and Dupoux 1996; Christophe et al. 2004; etc.)
  - Stress (Cutler and Norris 1988; Mattys and Samuel 1997)
  - Intonation (Kim 2004; Kim and Cho 2009)
  - Allophony (Gow and Gordon; less well studied)

Looking ahead

- Prosodic boundaries approach is based on the idea that phonology signals structure ⇒ parsing
- Next: two alternatives that derive boundary effects indirectly
  - Cyclic derivation: morpheme boundaries are string boundaries at some stage in the derivation
  - Output-output correspondence: segments at morpheme boundaries correspond to segments at word boundaries in related forms

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